Department of Information Systems

Bachelor of Information Systems

Lab 2

IS312 Web Application Development

Weighting: 4%
ULO addressed: ii, iii, iv
Due date: Monday 24th March 2025 @11:59PM
Total Marks: 30

HTML, CSS, JavaScript, PHP and MySQL (XAMPP)

Instructions:

1. Read the following problem carefully.
2. Submit your files (HTML, PHP, SQL) via the submission link on the unit page on Moodle before 11:59pm on Monday 24th March 2025.

Problem:

1. This Lab provides an opportunity for you to apply and integrate your knowledge and skills in HTML, CSS, JavaScript, PHP, SQL and MySQL.
2. Create a database called FRU10 in MySQL (XAMPP) with two tables, Student and Program, as shown and discussed in classes.
3. Create an HTML file (new-program.html), which will allow a user to enter new program details. When the user clicks on the submit button, the details are sent to the server (`add-newprogram.php`) using POST.
4. Create a PHP file called `add-new-program.php`. This file must contain PHP code that, when executed, will retrieve the form details, store them in variables and insert the details as a new record into the Program table in the database. If successful, display an appropriate message; otherwise an error message.
5. Create a PHP file called `student-listing.php`. This file must contain PHP code that, when executed, will retrieve all the student details from the database and display them on this page in a tabular format like the one shown below.

Table 1: Student Listing

StudentNo	Firstname	Lastname	Gender	ContactNo	*ProgramCode*
11111	James	Peter	Male	71717171	BS
22222	Peter	Mark	Male	71727172	IS
33333	Mary	John	Female	71737173	BS
44444	Belinda	Cain	Female	71717271	IS

6. Create an index page where the above functions can be accessed.
7. Ensure the quality of your code.
    a. Use a JavaScript function to validate the input controls to ensure that they are filled in. If a field is not filled in and the user clicks the Submit button, an appropriate message should be displayed to the user.
    b. Indent your code appropriately.
    c. Insert relevant comments as required and appropriately.
    d. At the top of each file, insert your name, class and date. For example:
    Author: Mr. Robin Pur
    Date: 22nd March 2025
    Unit: IS312 Web Application Development
8. From MySQL (XAMPP), export your database to an SQL file. To do so, click on your database and then click Export.

Marking Criteria

Criteria	Description	Marks
Database Creation (MySQL - XAMPP)	- Create a database named **FRU10** in MySQL. - Create **Student** and **Program** tables with correct structure and attributes.	3
HTML Form for Adding Programs (new-program.html)	- Form includes all required fields. - Uses POST method to send data to the server. - Proper HTML structure with labels and input fields.	3
3. PHP Script to Insert Program Data (add-new-program.php)	- Retrieves form data correctly using $_POST. - Stores data into appropriate variables. - Uses SQL INSERT statement to add a new record. - Displays a success or error message appropriately.	5
4. PHP Script for Student Listing (student-listing.php)	- Retrieves all student details from the database using SELECT. - Displays data in a well-structured HTML table. - Includes table headers and properly formatted output.	5
5. Index Page for Navigation	- Provides navigation links to all other functions (new-program.html, student-listing.php). - Has a clear and user-friendly interface.	2
6. JavaScript Form Validation	- Implements a JavaScript function to validate user input. - Ensures that required fields are filled before form submission. - Displays appropriate error messages if fields are empty.	4
7. Code Quality and Readability	- Proper code indentation. - Uses meaningful variable names. - Well-structured PHP and HTML code.	2
8. Comments and Documentation	- Each file contains author name, date, and unit details at the top. - Includes necessary comments explaining key sections of code.	2
9. Exported SQL File (Database Backup)	- Exports and submits the database as an SQL file. - The file contains all necessary tables and data.	2
	Total Marks	30